Extreme excursions and terminal-sign retention in stationary Brownian resetting

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Abstract

For stationary Brownian motion with independent exponential resetting, we determine the fixed-horizon probability of a high excursion followed by a positive terminal position. The exponential tail coefficient contains the probability of retaining a positive endpoint after the first reset, averaged over the remaining duration. An explicit Gaussian-tail remainder controls the finite-threshold approximation for every fixed positive horizon and reset rate and every finite reset position. This establishes the conjectured joint-tail asymptotic of Dębicki, Hashorva and Michna. When the reset position is zero, the coefficient is exactly midway between their conjectured bounds. The argument separates the stationary initial excursion from later reset-origin excursions and does not interchange terminal-threshold and high-excursion limits.

1 Observable and model

Endpoint monitoring of a resetting diffusion need not retain evidence of an earlier large excursion: a reset erases its position. We calculate this effect for a particle in a nonequilibrium stationary state on the full real line. Dębicki, Hashorva and Michna [1] obtained the stationary maximum tail and joint tails for terminal thresholds proportional to the excursion level. Their Remark 3 leaves the zero terminal threshold conjectural. We determine its coefficient and give an explicit finite-threshold error. Renewal and two-time methods for complete and incomplete resetting are developed in [2]; the contribution here is the stationary path-maximum/terminal-sign observable, rather than a new renewal formalism.

Let B be a standard Brownian motion, $\mathrm{Var}(B_t) = t$, and let N be an independent Poisson process of rate $r > 0$. Complete resets place the particle at $a \in \mathbb{R}$:

$$Y_t = Y_0 + B_t + \int_{(0,t]} (a - Y_{s-})\,\mathrm{d}N_s, \qquad p_\infty(y) = \frac{\alpha}{2} e^{-\alpha|y-a|}, \quad \alpha = \sqrt{2r}. \tag{1}$$

The initial position has density p_∞ and is independent of future noise. The diffusion coefficient is $D = 1/2$ in these units; there is no drift or reflecting boundary. The generator is $\mathcal{L}f(y) = f''(y)/2 + r[f(a) - f(y)]$. The displayed density solves $p_\infty''/2 - rp_\infty + r\delta_a = 0$ distributionally and has unit mass. Lengths are expressed in units for which the Brownian diffusion length is $\sqrt{t}$; thus αa, αu and $\alpha\sqrt{t}$ are dimensionless. Put $M_T = \sup_{0\le t\le T} Y_t$, with $T > 0$ fixed.

For a process started deterministically at a, its positive-terminal probability after duration $v > 0$ is

$$q_a(v) = e^{-rv}\Phi(a/\sqrt{v}) + \int_0^v re^{-rw}\Phi(a/\sqrt{w})\,\mathrm{d}w. \tag{2}$$

Here Φ is the standard normal distribution function. The last-reset age has an atom e^{-rv} at v and density re^{-rw} on $(0, v)$, proving (2). The strict event at zero duration gives $q_a(0) = \mathbf{1}_{\{a>0\}}$.

Its right limit is 1, 1/2 or 0 for positive, zero or negative a, respectively. Isolated endpoint values do not affect the integrals below.

2 Coefficient and finite-threshold bound

Theorem 1. *For every fixed $T > 0$, $r > 0$ and finite $a \in \mathbb{R}$ in* (1)*,*

$$\lim_{u\to\infty} e^{\alpha u}\mathbb{P}(M_T > u, Y_T > 0) = C(T,r,a) = e^{\alpha a}\left[\Phi(\alpha\sqrt{T}) + r\int_0^T \Phi(\alpha\sqrt{s})q_a(T-s)\,\mathrm{d}s\right]. \quad (3)$$

For $u > \max(a, 0)$, let $d = u - a$. Then

$$|e^{\alpha u}\mathbb{P}(M_T > u, Y_T > 0) - C(T,r,a)| \le R(u;T,r,a), \quad (4)$$

where

$$R(u;T,r,a) = e^{\alpha a + rT}\Phi\left(\frac{\alpha T - d}{\sqrt{T}}\right) + 2e^{\alpha u}\left[e^{-rT}\Phi\left(-\frac{u}{\sqrt{T}}\right) + rT\Phi\left(-\frac{d}{\sqrt{T}}\right)\right]. \quad (5)$$

The bound tends to zero for each fixed parameter triple.

In particular, C is finite and positive and obeys the strict inequalities

$$e^{\alpha a}\Phi(\alpha\sqrt{T}) < C(T,r,a) < e^{\alpha a}\left[\Phi(\alpha\sqrt{T}) + r\int_0^T \Phi(\alpha\sqrt{s})\,\mathrm{d}s\right]. \quad (6)$$

These establish the bounds conjectured in Remark 3 of [1]. No uniformity in simultaneous limits of T, r, a and u is asserted.

3 Proof

3.1 Separate the initial segment

Let τ be the first future reset and let E_u be a crossing of u in the initial Brownian segment, stopped at τ or T. Write $m_t = \sup_{0\le s\le t} B_s$ and

$$H(t,u) = \mathbb{P}(Y_0 + m_t > u), \qquad j(t,u) = \mathbb{P}(Y_0 + m_t > u, Y_0 + B_t > 0).$$

Conditional on $\tau = s$, complete resetting makes the future independent of the initial crossing, with terminal sign probability $q_a(T-s)$. Therefore

$$F(u) := \mathbb{P}(E_u, Y_T > 0) = e^{-rT}j(T,u) + \int_0^T re^{-rs}H(s,u)q_a(T-s)\,\mathrm{d}s. \quad (7)$$

Crossing at a reset instant has probability zero; continuity within the segment also represents its pre-reset supremum. Since $u > a$, a reset jump cannot create a crossing from below. Every crossing not counted in E_u must occur in one of the N_T later Brownian segments starting at a. Conditional on all reset times, each has length at most T. Reflection and a union bound give

$$0 \le \mathbb{P}(M_T > u, Y_T > 0) - F(u) \le 2rT\Phi(-d/\sqrt{T}). \quad (8)$$

Indeed the conditional sum is at most $2N_T\Phi(-d/\sqrt{T})$ and $\mathbb{E}N_T = rT$. Independence between segment events is unnecessary. This treats all reset counts without an infinite-series truncation.

3.2 Control loss at the Brownian endpoint

On $\{Y_0+m_T > u,\ Y_0+B_T \le 0\}$ one has $m_T - B_T > u$. The process $B_{T-h} - B_T$, $0 \le h \le T$, is a Brownian motion in distribution, so $m_T - B_T$ has the distribution of a Brownian maximum. For $u > 0$,

$$0 \le H(T,u) - j(T,u) \le 2\Phi(-u/\sqrt{T}). \tag{9}$$

The implication includes initial positions above u and does not impose a sign condition on the reset position.

3.3 Control the stationary initial tail

Let $X = Y_0 - a$ and $L(z) = \mathbb{P}(X > z)$. Its centered Laplace survival is

$$L(z) = \begin{cases} \frac{1}{2}e^{-\alpha z}, & z \ge 0, \\ 1 - \frac{1}{2}e^{\alpha z}, & z < 0. \end{cases}$$

For $d > 0$ and $m \ge 0$,

$$0 \le \frac{1}{2}e^{\alpha m} - e^{\alpha d}L(d-m) \le \frac{1}{2}e^{\alpha m}\mathbf{1}_{\{m>d\}}. \tag{10}$$

The difference is zero for $m \le d$; for $m > d$ its nonnegativity follows from $\cosh(\alpha(m-d)) \ge 1$, and its upper bound from $L \ge 0$. Since $H(t,u) = \mathbb{E}L(d - m_t)$ and $m_t \le m_T$ under the same Brownian path,

$$0 \le A(t) - e^{\alpha d}H(t,u) \le D_T(d), \qquad 0 \le t \le T, \tag{11}$$

where half-normal integration yields

$$\begin{aligned} A(t) &= \tfrac{1}{2}\mathbb{E}e^{\alpha m_t} = e^{rt}\Phi(\alpha\sqrt{t}), \\ D_T(d) &= \tfrac{1}{2}\mathbb{E}[e^{\alpha m_T}\mathbf{1}_{\{m_T>d\}}] = e^{rT}\Phi((\alpha T - d)/\sqrt{T}). \end{aligned} \tag{12}$$

At $t = 0$, $A(0) = 1/2$ and $H(0,u) = e^{-\alpha d}/2$, so the time endpoint is covered.

3.4 Assemble the bounds

In (7), replace $j(T,u)$ by $H(T,u)$ and then $e^{\alpha u}H(t,u)$ by $e^{\alpha a}A(t)$. Equation (12) gives exactly (3). The initial-tail error is at most $e^{\alpha a}D_T(d)$ times

$$e^{-rT} + \int_0^T re^{-rs}q_a(T-s)\,\mathrm{d}s \le 1.$$

The endpoint-loss error from (9) is at most $2e^{\alpha u - rT}\Phi(-u/\sqrt{T})$. The later-segment contribution in (8) adds $2rTe^{\alpha u}\Phi(-d/\sqrt{T})$. Their sum is (5). Each Gaussian tail vanishes after multiplication by a fixed exponential in u, proving (4) and the limit. Finally $0 < q_a(v) < 1$ for $v > 0$ and finite a, proving (6). □

4 Evaluation and physical interpretation

4.1 Controlled evaluation

Formula (3) involves only finite one-dimensional integrals. For exact-real enclosures, define $f_a(w) = \Phi(a/\sqrt{w})$ using its right limit at $w = 0$. On a partition $0 = w_0 < \cdots < w_n = v$, each contribution to the integral in (2) is between

$$(e^{-rw_i} - e^{-rw_{i+1}})\min\{f_a(w_i), f_a(w_{i+1})\} \quad \text{and the corresponding maximum.}$$

Monotonicity of f_a gives an enclosure width no greater than $r \max_i(w_{i+1} - w_i)$. Add the atom $e^{-rv} f_a(v)$. Differentiating (2) gives $q_a'(v) = e^{-rv} f_a'(v)$, so q_a is decreasing for $a > 0$, increasing for $a < 0$, and constant for $a = 0$. For the outer integral, endpoint bounds on both nonnegative factors bracket their product on every subinterval. Use the right limit of q_a at zero for these endpoint bounds. Refinement converges because the factors have bounded variation and continuous extensions from positive durations. Certified evaluations of Φ and the exponential, or directed rounding, are required to turn these exact-real bounds into machine certificates.

An alternative evaluation eliminates the inner quadrature. For $b = |a| > 0$, put

$$G(v) = e^{-\alpha b}\Phi(\alpha\sqrt{v} - b/\sqrt{v}) + e^{\alpha b}\Phi(-\alpha\sqrt{v} - b/\sqrt{v}). \tag{13}$$

Its derivative is $G'(v) = e^{-rv} b e^{-b^2/(2v)}/\sqrt{2\pi v^3}$ and $G(0+) = 0$. Consequently $q_a(v) = 1 - G(v)/2$ for $a > 0$ and $q_a(v) = G(v)/2$ for $a < 0$, by the derivative and initial limit of (2). Large exponential factors in (13) require stable special-function evaluation; the bounded-integral construction avoids that difficulty.

4.2 Zero reset position and retention

When $a = 0$, symmetry gives $q_0(v) = 1/2$ for $v > 0$:

$$C(T, r, 0) = \Phi(\alpha\sqrt{T}) + \frac{r}{2}\int_0^T \Phi(\alpha\sqrt{s})\,\mathrm{d}s. \tag{14}$$

This is the midpoint of the two bounding coefficients. With $k = \alpha\sqrt{T}$ and $\phi = \Phi'$,

$$\int_0^T \Phi(\alpha\sqrt{s})\,\mathrm{d}s = T\Phi(k) + \frac{k\phi(k) - \Phi(k) + 1/2}{\alpha^2}. \tag{15}$$

For $T = r = 1$, (14) gives $C \simeq 1.32857643$. This decimal is an ordinary floating-point evaluation of an exact expression.

The known unconditional coefficient in Theorem 5 of [1] is the upper expression in (6). Dividing C by that coefficient gives the limiting fraction of extreme paths retained by the positive endpoint monitor. The initial high segment controls the exponential scale; later excursions starting at a finite reset point have Gaussian-small crossing probabilities. After an initial excursion, a subsequent reset can erase the extreme position, but its remaining evolution retains the endpoint sign with probability q_a. The first term in (3) is the no-future-reset contribution; the integral quantifies this additional post-reset retention.

Changing a alters both the stationary tail multiplier and the post-reset sign probability, because the monitoring boundary stays at zero. Thus translation cannot be represented by the multiplier $e^{\alpha a}$ alone. The fixed zero threshold is evaluated directly: taking a terminal threshold proportional to u and then sending its proportionality factor to zero would not justify this result.

5 Scope and reproducibility

The result assumes a stationary start, zero drift, independent Poisson complete resets and a fixed finite horizon on the real line. Bound (5) may be loose at moderate thresholds, but vanishes at fixed parameters. It does not control joint parameter limits, other renewal laws or nonstationary starts. It is an exact model calculation, with no experimental inference.

The accompanying standard-library Python diagnostics evaluate the coefficient and compare last-reset and discounted first-hit representations of q_a. They also compare finite-threshold reflection/renewal calculations at several reset signs, horizons and rates. Mesh

refinement is diagnostic, not a certified discretization or roundoff bound; the analytic inequalities above establish the asymptotic and its error. Reproduction instructions and the environment description accompany the source files.

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Data availability. No experimental data were used. The calculation code and reproduction instructions are supplied with the local manuscript package.

References

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